

Sharp Fixed-Phase Order of Rvachev Quadrature

Dyadic maximality, effective refinement thresholds,
and quantitative defect geometry

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Abstract

A recent ProveIt manuscript classifies the phases at which the Rvachev lattice rule gains one polynomial degree, but leaves open whether a selected phase can gain a second degree. We settle this maximality question at every dyadic mesh. If $M = 2^d$, the largest degree attainable at any fixed phase is exactly $d + 1$. The two phases attaining it both fail at degree $d + 2$, with an explicit signed leading term and a relative error smaller than $1/4$.

For $M = 2^{r-1}m$, with m odd, we derive an exact next-defect formula and a computable sufficient condition for the same maximality conclusion. It holds eventually for every fixed m . More explicitly, if $J = \max\{2, 1 + \lceil \log_2 m \rceil\}$, it holds for every $r \geq 2J(m + 1) + 4$. Thus any exception at a fixed odd part lies in an explicitly bounded finite set of refinement levels. A refined expansion separates the 2^{-r} and 3^{-r} corrections and explains their parity dependence.

A three-factor Fourier estimate also shows that the first defect has exactly two nondegenerate stationary points, and gives a quantitative stability theorem for its zeros under C^1 perturbations. The results transfer to moment-deconvolved polynomial synthesis and to refined phase averages. The article supplies complete analytic proofs, independent exact and numerical checks, and a proposed formalization and research program.

Status. These are conventional mathematical proofs, not new Lean-verified theorems. The general all-integer-mesh maximality question is not claimed solved. Novelty is assessed relative to the inspected repository sources and primary references, not certified against the entire literature. This is an unrefereed research manuscript.

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1 The research question and the contribution

1.1 From phase completeness to fixed-phase maximality

Let u be the normalized Rvachev up-function: a nonnegative even C^∞ function supported on $[-1, 1]$, with integral one. Its Fourier transform, with the convention fixed below, is

$$H(t) = \widehat{u}(t) = \prod_{j=0}^{\infty} \operatorname{sinc}\left(\frac{\pi t}{2^j}\right), \quad \operatorname{sinc} z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases} \quad (1.1)$$

For $M > 0$ and a phase $\theta \in \mathbb{R}/\mathbb{Z}$, define

$$Q_{M,\theta}(P) = \frac{1}{M} \sum_{k \in \mathbb{Z}} P\left(\frac{k + \theta}{M}\right) u\left(\frac{k + \theta}{M}\right), \quad I(P) = \int_{\mathbb{R}} P(x) u(x) dx. \quad (1.2)$$

The actual spacing is $1/M$. The sum is finite. Write $\mathcal{P}_s$ for the polynomials of degree at most s , including zero.

For integer M , put $d = \nu_2(M)$, the exponent of two dividing M . The repository's module `RvachevSuperconvergentSynthesis.lean` establishes exactness on $\mathcal{P}_d$ at every phase, and sufficient special phases for exactness on $\mathcal{P}_{d+1}$ [4]. The subsequent manuscript *Complete Superconvergence Phases and Optimal Phase Filters for Rvachev Quadrature*, dated 28 September 2026, proves that these are all such phases [3]. Specifically, with $r = d + 1$,

$$Z_r = \begin{cases} \{0, \frac{1}{2}\} \pmod{1}, & r \text{ odd}, \\ \{\frac{1}{4}, \frac{3}{4}\} \pmod{1}, & r \text{ even}. \end{cases} \quad (1.3)$$

The same manuscript explicitly leaves the next-degree failure problem unresolved. Its results about optimal *uniformly translated* phase filters do not answer that different question.

Editorial note (ProveIt, 2026-09-29). This article answers Question 1 (“Fixed-phase maximality beyond the first extra degree”) of [3] for every dyadic mesh and, from the explicit level of Theorem 7.3 on, for every odd part; the finitely many lower levels of each non-dyadic mesh remain open. An editorial note at that question records this. The completeness of the phase set (1.3) at the first failing level is also Theorem `thm:phase-zero-set` of the canonical synthesis `inverse-and-sampling/comb-interpolation/comb_interpolation_synthesis/`, and the phase-uniform sharpness of exactness through degree $\nu_2(M)$ is machine-checked as `Fabius.exists_shift_tsum_shifted_monomial_ne_integral_nat_real` (`Analysis/FabiusFunction/Lean/FabiusFunction/CompositeMeshSharpness.lean`); neither is cited in this article. The fixed-phase results of this article have no Lean statement.

We ask whether, at a selected phase, the degree- $r + 1$ monomial still has a nonzero error. An affirmative answer at a mesh M completes its fixed-phase classification: all phases outside Z_r have exact degree $r - 1$, and the two phases in Z_r have exact degree r .

1.2 Main conclusions

The central result is Theorem 6.2: for every $d \geq 0$, the optimal fixed-phase degree at $M = 2^d$ is exactly $d + 1$. Its proof is quantitative, with a uniform relative margin; it is not an extrapolation from small meshes. The analytic mechanism is the explicit separation of the next Fourier defect into odd frequencies and a refined-mesh defect.

For a general odd part m , Theorem 6.1 gives an explicit sufficient inequality, Theorem 7.1 gives a cutoff in terms of two computable Fourier constants, and Theorem 7.3 gives the fully elementary bound

$$r \geq 2 \max\{2, 1 + \lceil \log_2 m \rceil\} (m + 1) + 4. \quad (1.4)$$

No unproved nonvanishing assumption occurs in that result. The constants are conservative, but the conclusion reduces each fixed-odd-part problem to finitely many levels, with an $O(m \log(m+1))$ explicit cutoff.

Two related conclusions are proved along the way. A strengthened odd-multiplier estimate gives the complete stationary-point geometry of the first defect and robust zero localization. The next-defect identity gives a two-exponential asymptotic expansion, with an entire even-frequency contribution disappearing at one of the two parities.

1.3 Scope, provenance, and limits

The repository snapshot inspected for the relevant source files was

23dd71d2d3d5e85d2b97a8cc45f55e735d69a4ae.

The pre-existing phase-completeness theorem, its two-factor estimate, the uniform filter classification, and classical dyadic evaluations are credited rather than presented as new. Primary descriptions of u and its arithmetic are due to Arias de Reyna [1, 2]. This article re-proves the few foundational facts needed to keep the new argument self-contained.

We have not audited the whole repository or independently certified the entire literature. In particular, the term “maximality” here concerns the fixed-node, fixed-weight family (1.2); it is not a statement about all Gaussian quadrature rules or arbitrary weights. We do not resolve every mesh $2^{r-1}m$ at the finitely many levels below our sufficient thresholds. The later research program states this remaining gap explicitly.

2 Fourier normalization and the first nonzero defect

2.1 Construction and zero multiplicities

Take independent variables V_j , each uniform on $[-1, 1]$, and let

$$Y = \sum_{j=1}^{\infty} 2^{-j} V_j.$$

The sum is absolutely convergent and belongs to $[-1, 1]$. Its characteristic function in the convention

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x) e^{-2\pi i t x} dx$$

is (1.1). The product converges locally uniformly on $\mathbb{C}$, since its tail factors differ from one by $O(4^{-j})$ on compact sets. It is an entire function.

For every fixed N , bounding the first N factors by reciprocal absolute arguments and the remaining real factors by one gives $H(t) = O_N(|t|^{-N})$. Fourier inversion therefore gives a smooth density u of Y . It is compactly supported, even, nonnegative, and normalized. Consequently H and all its real derivatives are Schwartz functions. This also justifies the classical normalization used in [1].

Lemma 2.1 (Real zeros). *The real zeros of H are exactly $\mathbb{Z} \setminus \{0\}$. Their orders are*

$$\text{ord}_N H = v_2(|N|) + 1 \quad (N \in \mathbb{Z} \setminus \{0\}). \quad (2.1)$$

In particular $H(m/2) \neq 0$ for odd m .

Proof. At a nonzero integer N of valuation d , precisely the factors with $j = 0, \dots, d$ vanish, each to first order. All the remaining factors are nonzero. Their tail product is nonzero because the sum of the absolute deviations from one converges. If t is not a nonzero integer, no factor vanishes, and the same tail argument excludes an additional product zero. $\square$

2.2 Poisson summation in physical coordinates

Set

$$E_{s,M}(\theta) = Q_{M,\theta}(x^s) - I(x^s). \quad (2.2)$$

The periodization of the Schwartz function $x^s u(x)$ gives

$$E_{s,M}(\theta) = \left(\frac{i}{2\pi}\right)^s \sum_{n \in \mathbb{Z} \setminus \{0\}} H^{(s)}(Mn) e^{2\pi i n \theta}. \quad (2.3)$$

For completeness, the n th Fourier coefficient of $M^{-1} \sum_k f((k + \theta)/M)$ is $\widehat{f}(Mn)$, by the substitution $x = (k + \theta)/M$. Differentiation under the integral gives $\widehat{x^s u}(t) = (i/(2\pi))^s H^{(s)}(t)$. Removing the constant coefficient proves (2.3). All phase derivatives of the series converge absolutely and uniformly.

From now on write

$$M = 2^{r-1}m, \quad r \geq 1, \quad m \geq 1 \text{ odd}. \quad (2.4)$$

By Lemma 2.1, $E_{s,M} \equiv 0$ for $s < r$. Thus only the degree- r and degree- $(r + 1)$ defects are needed for a sharp classification.

2.3 Normalized coefficients

Define the nonzero real number and real coefficients

$$A_m = H(m/2), \quad a_{m,n} = \frac{H(mn/2)}{A_m}, \quad b_{m,n} = \frac{m}{2} \frac{H'(mn/2)}{A_m} \quad (n \geq 1 \text{ odd}), \quad (2.5)$$

so $a_{m,1} = 1$. We often omit the subscript m inside proofs. Put

$$\beta_m = -b_{m,1} = -\frac{m}{2} \frac{H'(m/2)}{H(m/2)}, \quad \tau_s(\theta) = \cos(2\pi\theta + s\pi/2). \quad (2.6)$$

The letter $\mathcal{F}$ below denotes a normalized defect, not the Fabius function:

$$\mathcal{F}_{s,m}(\theta) = \sum_{\substack{n \geq 1 \\ n \text{ odd}}} \frac{a_{m,n}}{n^s} \tau_s(n\theta), \quad s \geq 1. \quad (2.7)$$

Its actual coefficients are rapidly decreasing for fixed m , so the series is smooth.

Proposition 2.2 (First defect; established phase framework). *For the mesh (2.4),*

$$E_{r,M}(\theta) = -\frac{2r!A_m}{(2\pi M)^r} \mathcal{F}_{r,m}(\theta). \quad (2.8)$$

Proof. At $N = Mn$ with n odd, split the product after its first r factors. Exactly these factors vanish, and their first-order product is $-(z - N)^r / N^r$: among their integer arguments $N/2^j$, exactly the last is odd. Hence

$$H^{(r)}(Mn) = -\frac{r!}{(Mn)^r} H(mn/2).$$

For even n , the derivative vanishes. Pair positive and negative odd frequencies in (2.3), using $H^{(r)}(-t) = (-1)^r H^{(r)}(t)$, to obtain (2.8). $\square$

3 A strengthened multiplier estimate and defect geometry

Write

$$S_s = \sum_{\substack{n \geq 3 \\ n \text{ odd}}} n^{-s} = (1 - 2^{-s})\zeta(s) - 1 \quad (s > 1). \quad (3.1)$$

The following elementary bound will be useful:

$$S_s \leq 3^{-s} + \frac{1}{2} \int_3^\infty x^{-s} dx = 3^{-s} \left(1 + \frac{3}{2(s-1)} \right). \quad (3.2)$$

Indeed, for every odd $n \geq 5$, the integral over $[n-2, n]$ is at least $2n^{-s}$.

Lemma 3.1 (Two-factor and three-factor domination). *For positive odd m, n ,*

$$|a_{m,n}| \leq \min \left\{ \frac{1}{n^2}, \frac{1 + \sqrt{2}}{n^3} \right\}. \quad (3.3)$$

Proof. For integer $n \geq 1$ and real x , $|\sin(nx)| \leq n|\sin x|$. One proof expands $\sin(nx)/\sin x$ as a sum of n complex numbers of modulus one when the denominator is nonzero, and then uses continuity. Thus every individual sinc-factor ratio in $H(mn/2)/H(m/2)$ has absolute value at most one.

For $j = 0$, the arguments are odd multiples of $\pi/2$, so the sine magnitudes are both one and the ratio is exactly $1/n$. For $j = 1$, the sine magnitudes at odd multiples of $\pi/4$ are both $1/\sqrt{2}$, so that ratio is also $1/n$. This proves the first bound, the estimate used in the earlier phase-completeness proof [3].

At $j = 2$, the sine magnitudes at odd multiples of $\pi/8$ belong to $\{\sin(\pi/8), \sin(3\pi/8)\}$. Their ratio is at most $\sin(3\pi/8)/\sin(\pi/8) = 1 + \sqrt{2}$. That factor contributes at most $(1 + \sqrt{2})/n$, and all later ratios are at most one. Passing from finite products to their nonzero limits proves the second bound. $\square$

Theorem 3.2 (Two zeros and exactly two stationary points). *Put $c_r = S_{r+1}$ and $\eta_r = (1 + \sqrt{2})S_{r+1}$. Then, for every odd m and $r \geq 1$,*

$$|\mathcal{F}_{r,m}(\theta) - \tau_r(\theta)| \leq c_r |\tau_r(\theta)|, \quad (3.4)$$

$$|\mathcal{F}'_{r,m}(\theta) - \tau'_r(\theta)| \leq \eta_r |\tau'_r(\theta)|. \quad (3.5)$$

Here $c_r \leq \pi^2/8 - 1 < 1/4$ and $\eta_r < 1$. The zeros of $\mathcal{F}_{r,m}$ are exactly Z_r and are simple. Its stationary points are exactly the zeros of τ'_r , and are nondegenerate. It has the same sign and the same monotonicity intervals as τ_r .

Proof. For odd n , the integer multiple inequality, applied to either sine or cosine according to the parity of r , gives

$$|\tau_r(n\theta)| \leq n|\tau_r(\theta)|.$$

Combining this with the first bound in (3.3) proves (3.4). This is the known phase-completeness argument, reproduced here to establish the exact baseline.

Differentiation contributes a factor n , and the same trigonometric inequality applied to the differentiated harmonic gives

$$\left| \frac{d}{d\theta} \tau_r(n\theta) \right| \leq n^2 |\tau'_r(\theta)|.$$

The three-factor bound therefore proves (3.5). Since $S_{r+1} \leq S_2 = \pi^2/8 - 1 < 1/4$ and $1 + \sqrt{2} < 5/2$, one even has $\eta_r < 5/8 < 1$; its numerical upper value is about 0.5642.

The first inequality identifies the zero set and sign; the second identifies the stationary set and sign of the derivative. At a zero of τ_r , its derivative is nonzero, so (3.5) gives simplicity. At a stationary point θ_0 , divide (3.5) by $|\theta - \theta_0|$ and take the limit. The resulting bound

$$|\mathcal{F}_{r,m}''(\theta_0) - \tau_r''(\theta_0)| \leq \eta_r |\tau_r''(\theta_0)|$$

shows that the second derivative is nonzero there. $\square$

Corollary 3.3 (Baseline exactness classification). *At the mesh (2.4), the rule is exact on $\mathcal{P}_r$ if and only if $\theta \in Z_r$. At every other phase its exact degree is $r - 1$.*

Proof. All lower monomial defects vanish. The remaining degree- r defect is a nonzero scalar times (2.7), and Theorem 3.2 locates its zeros. $\square$

3.1 Quantitative stability under perturbation

The relative derivative estimate excludes an instability that a uniform function bound alone would miss: a very small, highly oscillatory perturbation can introduce many zeros unless its derivative is also controlled.

Theorem 3.4 (Stable two-zero structure). *Fix $0 < \varepsilon < 1/4$, and let g be a real C^1 function on $\mathbb{R}/\mathbb{Z}$. Suppose*

$$\|g\|_\infty < (1 - c_r) \sin(2\pi\varepsilon), \quad (3.6)$$

$$\|g'\|_\infty < 2\pi(1 - \eta_r) \cos(2\pi\varepsilon). \quad (3.7)$$

Then $\mathcal{F}_{r,m} + g$ has exactly two zeros. Each lies in the ε -neighborhood of a different point $z \in Z_r$, and its circular distance from that point is at most

$$\frac{\|g\|_\infty}{2\pi(1 - \eta_r) \cos(2\pi\varepsilon)}. \quad (3.8)$$

Both zeros are simple.

Proof. Outside the two neighborhoods, $|\tau_r| \geq \sin(2\pi\varepsilon)$. Equation (3.4) and (3.6) exclude a zero there. At the two endpoints of each neighborhood, the unperturbed function has opposite signs, and those signs survive the perturbation. Hence a zero exists in each neighborhood.

Inside such a neighborhood, $|\tau_r'| \geq 2\pi \cos(2\pi\varepsilon)$, with constant sign. Equations (3.5) and (3.7) imply that the perturbed derivative is nonzero with that sign throughout. This gives uniqueness and simplicity. Apply the mean value theorem to the unperturbed function between z and its perturbed zero to obtain (3.8). $\square$

4 An exact formula for the next defect

Set $p = r + 1$. The first defect sees only odd frequencies; the next defect sees odd frequencies and frequencies of valuation one. Treating these two sets separately is the main structural step.

Lemma 4.1 (The next Taylor coefficient at an odd alias). *If $N = Mn$ and n is odd, then*

$$H^{(r+1)}(N) = \frac{(r+1)!}{N^r} \left(\frac{r}{N} H(mn/2) - 2^{-r} H'(mn/2) \right). \quad (4.1)$$

Proof. Write $z = N + w$. For each $j < r$,

$$\operatorname{sinc}\left(\frac{\pi(N+w)}{2^j}\right) = (-1)^{N/2^j} \frac{w}{N} \left(1 - \frac{w}{N} + O(w^2)\right).$$

The sine numerator has no quadratic term in w ; the displayed linear correction comes from the reciprocal denominator. Multiplying the first r factors, whose signs have product -1 , gives

$$-\frac{w^r}{N^r} \left(1 - \frac{rw}{N} + O(w^2)\right).$$

The remaining product is $H(z/2^r) = H(mn/2) + 2^{-r} w H'(mn/2) + O(w^2)$. Taking the coefficient of w^{r+1} and multiplying by $(r+1)!$ proves the formula. All expansions are justified by analyticity near N . $\square$

Theorem 4.2 (Odd aliases plus a refined first defect). *Define the signed, nonzero normalization*

$$K_{r,m} = \frac{2(r+1)!A_m}{(2\pi M)^{r+1}}. \quad (4.2)$$

For every $\theta \in \mathbb{R}/\mathbb{Z}$,

$$\boxed{\frac{E_{r+1,M}(\theta)}{K_{r,m}} = \sum_{\substack{n \geq 1 \\ n \text{ odd}}} \left(\frac{ra_{m,n}}{n^{r+1}} - \frac{b_{m,n}}{n^r} \right) \tau_{r+1}(n\theta) - 2^{-(r+1)} \mathcal{F}_{r+1,m}(2\theta)}. \quad (4.3)$$

The first term of the odd sum is $(r + \beta_m)\tau_{r+1}(\theta)$.

Proof. Pairing the positive and negative odd frequencies in (2.3), and using Lemma 4.1, gives

$$\frac{2(r+1)!A_m}{(2\pi M)^{r+1}} \sum_{\substack{n \geq 1 \\ n \text{ odd}}} \left(\frac{ra_{m,n}}{n^{r+1}} - \frac{b_{m,n}}{n^r} \right) \tau_{r+1}(n\theta),$$

because $M2^{-r} = m/2$. The entire even-frequency part is exactly $E_{r+1,2M}(2\theta)$, by replacing n with $2n$ in the Fourier series. The mesh $2M = 2^r m$ has first threshold $r+1$, so Proposition 2.2 expresses that even part as $-2^{-(r+1)} K_{r,m} \mathcal{F}_{r+1,m}(2\theta)$. Finally $a_{m,1} = 1$ and $b_{m,1} = -\beta_m$. $\square$

Corollary 4.3 (Parity cancellation of the even component). *At a selected phase $\theta \in Z_r$, the even component in (4.3) vanishes identically when r is even. When r is odd, 2θ is an integer and*

$$|\mathcal{F}_{r+1,m}(2\theta)| \leq 1 + S_{r+3}. \quad (4.4)$$

Proof. If r is even, $\theta \in \{1/4, 3/4\}$ and $r+1$ is odd. Every harmonic of $\mathcal{F}_{r+1,m}$ is a sine, hence vanishes at $2\theta \in \{1/2, 3/2\}$. For odd r , use $2\theta \in \mathbb{Z}$ and $|a_{m,n}| \leq n^{-2}$ in the series. $\square$

5 Derivative domination with explicit constants

The next defect involves derivatives of H at all odd half-integer multiples, not only its values. Define

$$P(t) = \operatorname{sinc}(\pi t) \operatorname{sinc}(\pi t/2), \quad T(t) = \prod_{j=2}^{\infty} \operatorname{sinc}(\pi t/2^j), \quad H = PT. \quad (5.1)$$

The tail T is the Fourier transform of a probability supported on $[-1/4, 1/4]$, so

$$|T'(t)| \leq \frac{\pi}{2} \quad (t \in \mathbb{R}). \quad (5.2)$$

Indeed, its random-variable representation is the part of Y left after removing its first two summands, and differentiation contributes at most 2π times its support radius.

Lemma 5.1 (Uniform derivative envelope along odd multiples). *For a positive odd m , let*

$$D_m = \frac{m\pi}{4} \left(1 + \frac{1}{|T(m/2)|} \right) + \frac{2}{3}. \quad (5.3)$$

This is finite, and for every odd $n \geq 3$,

$$|b_{m,n}| \leq \frac{D_m}{n^2}. \quad (5.4)$$

Proof. Put $t_0 = m/2$. All the factors at t_0 are nonzero. The two exact factor ratios from Lemma 3.1 give $|P(nt_0)| = |P(t_0)|/n^2$. The same factorwise inequality gives $|T(nt_0)| \leq |T(t_0)|$.

At an odd half-integer,

$$\frac{P'(mn/2)}{P(mn/2)} = \frac{\pi}{2} \cot\left(\frac{\pi mn}{4}\right) - \frac{4}{mn}.$$

The missing first cotangent is zero. The displayed cotangent has magnitude one, hence the whole expression has magnitude at most $\pi/2 + 4/(mn)$. Differentiating $H = PT$ and applying (5.2) yields

$$\frac{|H'(mn/2)|}{|A_m|} \leq \frac{1}{n^2} \left(\frac{\pi}{2} + \frac{4}{mn} + \frac{\pi}{2|T(m/2)|} \right).$$

Multiply by $m/2$ and use $2/n \leq 2/3$. □

Lemma 5.2 (The dyadic constants have absolute bounds). *For $m = 1$,*

$$A_1 > 0, \quad \beta_1 > 1, \quad D_1 < \frac{5}{2}. \quad (5.5)$$

Proof. Every factor of $H(1/2)$ is positive. Each sinc factor is strictly decreasing as a function of its positive argument in $(0, \pi/2]$. The $j = 0$ logarithmic derivative at $t = 1/2$ is -2 and the remaining logarithmic derivatives are negative. Therefore $\beta_1 = -\frac{1}{2}H'(1/2)/H(1/2) > 1$.

For the tail, sinc $x \geq 1 - x^2/6$ for real x , and its arguments here are positive and at most $\pi/8$. The elementary product inequality $1 - \prod_j (1 - v_j) \leq \sum_j v_j$ for $0 \leq v_j \leq 1$ gives

$$T(1/2) \geq 1 - \sum_{j=2}^{\infty} \frac{\pi^2}{24 \cdot 4^j} = 1 - \frac{\pi^2}{288} > \frac{24}{25}.$$

For the final inequality, $\pi < 22/7$ is more than sufficient. Hence

$$D_1 < \frac{11}{14} \left(1 + \frac{25}{24} \right) + \frac{2}{3} = \frac{763}{336} < \frac{5}{2}.$$

□

Remark 5.3 (The odd part cannot be discarded). The value β_m need not be positive. Numerically, $\beta_9 \approx -10.8871$ and $\beta_{17} \approx -40.7636$. Thus the leading coefficient $r + \beta_m$ can be small near particular levels. The proof for $m = 1$ must not be applied to all m by silently assuming $\beta_m > 1$. The general criterion below uses its actual sign and magnitude.

6 Sharp fixed-phase order at every dyadic mesh

The following general error envelope is the bridge from Fourier analysis to a nonvanishing theorem. Define

$$R_{r,m} = rS_{r+3} + D_m S_{r+2} + \mathbf{1}_{\{r \text{ odd}\}} 2^{-(r+1)} (1 + S_{r+3}). \quad (6.1)$$

The indicator retains the exact cancellation of the even-frequency part.

Theorem 6.1 (A quantitative maximality criterion). *For every $\theta \in Z_r$,*

$$\left| \frac{E_{r+1,M}(\theta)}{K_{r,m}} - (r + \beta_m)\tau_{r+1}(\theta) \right| \leq R_{r,m}. \quad (6.2)$$

Consequently, if

$$|r + \beta_m| > R_{r,m}, \quad (6.3)$$

then the exact degree of $Q_{M,\theta}$ is r at each $\theta \in Z_r$ and $r - 1$ at every other phase. In particular, its maximum over all phases is r . At a selected phase,

$$\begin{aligned} |K_{r,m}|(|r + \beta_m| - R_{r,m}) &\leq |E_{r+1,M}(\theta)| \\ &\leq |K_{r,m}|(|r + \beta_m| + R_{r,m}), \end{aligned} \quad (6.4)$$

and its sign is that of $K_{r,m}(r + \beta_m)\tau_{r+1}(\theta)$.

Proof. At a selected phase, $|\tau_{r+1}(\theta)| = 1$. Remove the $n = 1$ term from the odd sum in Theorem 4.2. Its remaining magnitude is at most

$$\sum_{\substack{n \geq 3 \\ n \text{ odd}}} \left(\frac{r}{n^{r+3}} + \frac{D_m}{n^{r+2}} \right) = rS_{r+3} + D_mS_{r+2},$$

by Lemmas 3.1 and 5.1. Add the even-component bound from Corollary 4.3 to obtain (6.2). Under (6.3), the leading term cannot be canceled. The two-sided bound and sign follow, and Corollary 3.3 completes the degree classification. $\square$

Theorem 6.2 (Complete dyadic fixed-phase maximality). *For every $d \geq 0$, put $M = 2^d$ and $r = d + 1$. Then*

$$\deg_{\text{exact}} Q_{2^d,\theta} = \begin{cases} d + 1, & \theta \in Z_{d+1}, \\ d, & \theta \notin Z_{d+1}. \end{cases} \quad (6.5)$$

Here “exact degree” means the largest s such that the rule is exact on all of $\mathcal{P}_s$. More quantitatively, at either selected phase,

$$\left| \frac{E_{r+1,2^{r-1}}(\theta)}{K_{r,1}(r + \beta_1)\tau_{r+1}(\theta)} - 1 \right| < \frac{1}{4}. \quad (6.6)$$

In particular, both selected phases fail at the very next degree.

Proof. By Lemma 5.2, $r + \beta_1 > r + 1$ and $D_1 < 5/2$. We may ignore the parity indicator for an upper bound. Therefore

$$\frac{R_{r,1}}{r + \beta_1} < \frac{rS_{r+3} + (5/2)S_{r+2} + 2^{-(r+1)}(1 + S_{r+3})}{r + 1}.$$

All terms are uniformly controlled by the first refinement level. For $r \geq 1$,

$$\frac{rS_{r+3}}{r + 1} \leq \frac{1}{2}S_4, \quad \frac{S_{r+2}}{r + 1} \leq \frac{1}{2}S_3, \quad \frac{2^{-(r+1)}(1 + S_{r+3})}{r + 1} \leq \frac{1}{8}(1 + S_4).$$

For the first inequality, use $S_{r+3} \leq 3^{-(r-1)}S_4$ and $r/(r + 1) \leq 3^{r-1}/2$; the latter follows at $r = 1$ by equality and for $r \geq 2$ from $r/(r + 1) < 1 \leq 3^{r-1}/2$. The other inequalities follow directly by monotonicity.

Equation (3.2) gives $S_3 \leq 7/108$ and $S_4 \leq 1/54$. Consequently

$$\frac{R_{r,1}}{r + \beta_1} < \frac{1}{2}S_4 + \frac{5}{4}S_3 + \frac{1}{8}(1 + S_4) \leq \frac{47}{216} < \frac{1}{4}. \quad (6.7)$$

Apply Theorem 6.1; division by the signed leading term gives (6.6). The error cannot be zero, so the selected phases have exact degree r . All other phases were already classified. $\square$

6.1 Why the result is stronger than phase completeness

Knowing that the degree- r error has exactly two zeros does not exclude one of those zeros from also annihilating the degree- $(r + 1)$ error. The new ingredient is (6.6): it separates the next error from zero at *both* selected phases, uniformly over all d . The proof uses more than symmetry. Symmetry supplies selected zeros, but only the derivative envelope and the refined-mesh decomposition control the next error there.

6.2 Signs and the two selected phases

Since $A_1 > 0$ and $r + \beta_1 > 0$, the sign of the next dyadic defect is precisely $\tau_{r+1}(\theta)$. At the pair of selected phases these signs are opposite. Their magnitudes need not be equal when r is odd: the even-frequency component is then identical at the two phases and shifts both errors by the same amount. When r is even the errors are exact negatives, because the even component vanishes and all remaining frequencies are odd.

For example, the rule at $M = 1$, $\theta = 0$ has only the central nonzero sample and gives $Q(x^2) = 0$, whereas $I(x^2) = 1/9$. At $\theta = 1/2$, the two samples at $\pm 1/2$ give $Q(x^2) = 1/4$, and the error is $5/36$. This simple asymmetric pair is the first instance of the parity mechanism, not an exceptional normalization artifact.

7 Every fixed odd part has an effective finite frontier

7.1 A cutoff from computable Fourier constants

Theorem 7.1 (Eventual maximality at every fixed odd part). *For positive odd m , define the integer*

$$r_*(m) = \max \{2, \lceil 2|\beta_m| + 2 \rceil, 2 + \lceil \log_3(1 + D_m) \rceil\}. \quad (7.1)$$

For every integer $r \geq r_(m)$, the mesh $M = 2^{r-1}m$ has exact fixed-phase degree r on Z_r and $r - 1$ off Z_r . In particular, any failure of this classification for a fixed odd m occurs at one of the finitely many levels $1 \leq r < r_*(m)$.*

Proof. For $s \geq 3$, (3.2) gives $S_s \leq 2 \cdot 3^{-s}$. Moreover, $S_{r+3} < 1$, so

$$R_{r,m} \leq \frac{2r}{3^{r+3}} + \frac{2D_m}{3^{r+2}} + 2^{-r}. \quad (7.2)$$

Under (7.1), $D_m < 3^{r-2}$, $r + \beta_m \geq r - |\beta_m| \geq r/2 + 1$, and $r \geq 2$. The first term in (7.2) is at most $4/243$, since $r/3^r$ decreases for integer $r \geq 2$. The second is less than $2/81$ and the third at most $1/4$. Their sum is less than one, while the leading coefficient is greater than one. Hence (6.3) holds, and Theorem 6.1 applies. $\square$

The cutoff is intentionally sufficient, not optimal. It can be much larger than the first level at which (6.3) succeeds. A failed sufficient test is not evidence that the underlying error is zero.

7.2 A fully elementary cutoff, without special-function constants

To remove even the need to evaluate β_m and D_m , we bound them using dyadic separation from the sine zeros.

Lemma 7.2 (Elementary bounds on the odd-part constants). *For odd $m \geq 1$ let*

$$J = \max\{2, 1 + \lceil \log_2 m \rceil\}. \quad (7.3)$$

Then

$$|\beta_m| < J(m+1) + 1, \quad D_m < m + (2m)^{J-1} + 1 \leq (2m)^J. \quad (7.4)$$

Proof. Set $t = m/2$ and $c_j = \pi/2^j$. Since $2^J \geq 2m$ and $\pi < 4$, $|c_j t| < 1$. At an odd half-integer the angle $c_j t = \pi m/2^{j+1}$ has distance at least $\pi/2^{j+1}$ from the nearest integer multiple of π . Since $|\cot x| \leq 1/\delta$ when that distance is $\delta \in (0, \pi/2]$,

$$|c_j \cot(c_j t)| \leq 2, \quad |c_j \cot(c_j t) - 1/t| \leq 2 + 2/m.$$

Thus the first J contributions to $|tH'(t)/H(t)|$ total at most $J(m+1)$.

For $|x| \leq 1$, the bounds $|\sin x| \geq 5|x|/6$ and $|x \cos x - \sin x| \leq |x|^3/3$ give

$$|x \cot x - 1| \leq \frac{2}{5}x^2. \quad (7.5)$$

The numerator bound follows by integrating $-x \sin x$ from zero. Consequently the logarithmic-derivative tail contributes at most

$$\frac{2}{5}t^2 \sum_{j=J}^{\infty} c_j^2 = \frac{8\pi^2 t^2}{15} 4^{-J} = \frac{2\pi^2 m^2}{15} 4^{-J} < \frac{8}{15} < 1.$$

This proves the bound for $|\beta_m|$.

For the bound on D_m , every factor of $T(t)$ with $2 \leq j < J$ has magnitude at least $2/(\pi m) > 1/(2m)$. Indeed, its sine numerator has magnitude at least $\sin(\pi/2^{j+1}) \geq 2^{-j}$, whereas its argument has magnitude $\pi m/2^{j+1}$. For $j \geq J$ all factors are positive. The same product estimate used in Lemma 5.2 gives

$$\prod_{j=J}^{\infty} \text{sinc}(c_j t) \geq 1 - \frac{2\pi^2 t^2}{9} 4^{-J} = 1 - \frac{\pi^2 m^2}{18} 4^{-J} > \frac{7}{9}.$$

It follows that

$$|T(m/2)| > \frac{7}{9}(2m)^{-(J-2)}.$$

Using $\pi/4 < 1$, $9/7 < 2$, and $2/3 < 1$ in (5.3) gives

$$D_m < m(1 + 2(2m)^{J-2}) + 1 = m + (2m)^{J-1} + 1.$$

The last expression is at most $(2m)^J$: since $J \geq 2$, $(2m)^{J-1}(2m-1) \geq 2m(2m-1) \geq m+1$ for $m \geq 1$. $\square$

Theorem 7.3 (Explicit finite-level reduction). *For every odd $m \geq 1$, with J as in (7.3), all refinement levels*

$$r \geq 2J(m+1) + 4 \quad (7.6)$$

have the sharp fixed-phase classification of Theorem 6.1. Equivalently, at fixed odd part m , it suffices to settle the finitely many levels

$$1 \leq r \leq 2J(m+1) + 3 \quad (7.7)$$

to complete the all-level classification for that odd part.

Proof. The β_m bound in Lemma 7.2 makes $2|\beta_m| + 2 < 2J(m+1) + 4$. Also

$$1 + D_m < 1 + (2m)^J \leq 2(2m)^J \leq 2 \cdot 3^{Jm} < 3^{Jm+1},$$

where $2m \leq 3^m$ for integers $m \geq 1$. Thus $2 + \lceil \log_3(1 + D_m) \rceil \leq Jm + 3$, which is smaller than the right side of (7.6). The cutoff in Theorem 7.1 is therefore satisfied. $\square$

This reduction is finite for each prescribed m , not uniformly finite over all odd integers. The possible number of unresolved levels still grows with m . In particular, Theorem 7.3 is not a proof that there are only finitely many exceptional integer meshes altogether.

7.3 Effective evaluation and rigorous numerical certificates

The constants in (7.1) are computable with certified errors. For $t > 0$, choose J so that $\pi t 2^{-J} \leq 1$. The product tail has

$$0 \leq 1 - \prod_{j=J}^{\infty} \operatorname{sinc}(\pi t / 2^j) \leq \frac{2\pi^2 t^2}{9} 4^{-J}. \quad (7.8)$$

The logarithmic-derivative tail obeys, by (7.5),

$$\left| \sum_{j=J}^{\infty} \left(\frac{\pi}{2^j} \cot\left(\frac{\pi t}{2^j}\right) - \frac{1}{t} \right) \right| \leq \frac{8\pi^2 t}{15} 4^{-J}. \quad (7.9)$$

At an odd half-integer, the finite initial product is nonzero. These tail bounds therefore provide a positive lower bound on its absolute value once the cutoff is sufficiently large. Interval bounds on the finite sines, cosines, and π then enclose β_m and D_m .

There is no algorithmic need to decide whether a transcendental expression is an integer before choosing a cutoff: rational upper enclosures $B > |\beta_m|$ and $D > D_m$ give a conservative integer cutoff by rounding up. The completely elementary cutoff is available without these computations. The supplied numerical implementation illustrates the sharper constants, but does not implement directed-rounding intervals; its output is explicitly labeled diagnostic.

8 The two-exponential structure of the next error

The exact formula contains more information than eventual nonvanishing. It distinguishes the first even alias from the first nontrivial odd alias, and shows why their relative importance depends on parity.

For $s > 1$ write

$$U_s = \sum_{\substack{n \geq 5 \\ n \text{ odd}}} n^{-s}.$$

As in (3.2),

$$U_s \leq 5^{-s} \left(1 + \frac{5}{2(s-1)} \right). \quad (8.1)$$

Theorem 8.1 (Explicit three-term expansion at the selected phases). *For $\theta \in Z_r$ and fixed odd m ,*

$$\begin{aligned} \frac{E_{r+1,M}(\theta)}{K_{r,m}} &= (r + \beta_m) \tau_{r+1}(\theta) - 2^{-(r+1)} \tau_{r+1}(2\theta) \\ &\quad + 3^{-r} \left(\frac{r}{3} a_{m,3} - b_{m,3} \right) \tau_{r+1}(3\theta) + \mathcal{R}_{r,m}(\theta), \end{aligned} \quad (8.2)$$

where

$$|\mathcal{R}_{r,m}(\theta)| \leq r U_{r+3} + D_m U_{r+2} + \mathbf{1}_{\{r \text{ odd}\}} 2^{-(r+1)} S_{r+3}. \quad (8.3)$$

Thus, as $r \rightarrow \infty$ with m fixed, the remainder is $O_m(r 5^{-r} + 6^{-r})$, uniformly over the two selected phases. When r is even, every even-alias term, including its remainder, is exactly zero.

Proof. In the odd part of (4.3), retain $n = 1$ and $n = 3$. Their contributions are exactly the first and third terms of (8.2). The remaining odd terms are bounded by $r U_{r+3} + D_m U_{r+2}$. In the refined term retain only its $n = 1$ harmonic. Its remaining terms have magnitude at most $2^{-(r+1)} S_{r+3}$, unless r is even, when all its harmonics vanish by Corollary 4.3. Finally use (8.1) and (3.2). $\square$

Corollary 8.2 (Signed leading asymptotic). *For fixed odd m , uniformly over $\theta \in \mathbb{Z}_r$,*

$$E_{r+1,2^{r-1}m}(\theta) = K_{r,m}(r + \beta_m)\tau_{r+1}(\theta) \left(1 + O_m \left(3^{-r} + \frac{2^{-r}}{r} \right) \right). \quad (8.4)$$

On the even- r subsequence the term $2^{-r}/r$ may be omitted.

Proof. For fixed m , $r + \beta_m \sim r$. Divide the error estimate in Theorem 6.1 by $|r + \beta_m|$, and use (3.2). The even- r improvement follows from the exact cancellation. $\square$

The two small parameters are not interchangeable. For odd r , the 2^{-r} term occurs before the $r3^{-r}$ odd correction. For even r , the whole 2^{-r} family disappears. This is a simple instance of an exponential asymptotic scale selected by arithmetic parity, directly relevant to the repository's broader interest in transseries. No divergent series, summability prescription, or beyond-all-orders assertion is needed here: the finite truncation has an explicit remainder bound.

8.1 Exact sum and difference of the two errors

Let θ_0 and $\theta_0 + 1/2$ be the selected phases. Every odd harmonic changes sign under this shift, whereas the refined component is unchanged. Hence the identity

$$E_{r+1,M}(\theta_0) + E_{r+1,M}(\theta_0 + 1/2) = 2E_{r+1,2M}(2\theta_0) \quad (8.5)$$

is exact. Their difference isolates twice the odd part of (4.3). In particular, if r is even their sum is zero. If r is odd, the sum has the sign and size of a first defect on the refined mesh. This supplies a direct check on both the factor $2^{-(r+1)}$ and its minus sign in the main formula.

9 Polynomial synthesis and refined phase filters

9.1 Moment deconvolution preserves the sharp degree

Define the polynomial averaging operator

$$(\mathcal{B}P)(t) = \int_{\mathbb{R}} P(t-x)u(x) \, dx. \quad (9.1)$$

In the monomial basis it is triangular with diagonal one, so it has an inverse $\mathcal{D}$ on the polynomial algebra. Both operators preserve degree and leading coefficient. The polynomials $A_n = \mathcal{D}(t^n)$ are monic and have the formal generating identity

$$\sum_{n \geq 0} A_n(t) \frac{z^n}{n!} = \frac{e^{tz}}{\int_{\mathbb{R}} e^{zx} u(x) \, dx}.$$

Evenness of u identifies its moment-generating function at z and $-z$. Only a formal power series, or a neighborhood of zero, is intended; the reciprocal denominator is not asserted to be entire.

The phased synthesis operator is

$$(\mathcal{S}_{M,\theta}P)(t) = \frac{1}{M} \sum_{k \in \mathbb{Z}} (\mathcal{D}P) \left(t - \frac{k+\theta}{M} \right) u \left(\frac{k+\theta}{M} \right). \quad (9.2)$$

This is the deconvolution framework already present in the repository [4, 3].

Theorem 9.1 (Sharp synthesis order and its leading failure). *The identity $\mathcal{S}_{M,\theta}P = P$ for every $P \in \mathcal{P}_s$ is equivalent to exactness of $\mathcal{Q}_{M,\theta}$ on $\mathcal{P}_s$. If the quadrature is exact through degree r and $p = r + 1$, then*

$$(\mathcal{S}_{M,\theta}t^p)(t) - t^p = (-1)^p E_{p,M}(\theta), \quad (9.3)$$

independently of t . More generally the error for a degree-at-most- p polynomial is its leading degree- p coefficient times the right side. Consequently all the sharp-order conclusions of Theorems 6.2, 7.1 and 7.3 apply to phased synthesis.

Proof. Quadrature exactness applies to the polynomial $x \mapsto (\mathcal{D}P)(t-x)$, whose integral is $(\mathcal{B}\mathcal{D}P)(t) = P(t)$. This proves one direction. Conversely, as P ranges over $\mathcal{P}_s$, the polynomials $x \mapsto (\mathcal{D}P)(-x)$ range over all of $\mathcal{P}_s$, since $\mathcal{D}$ and reflection are invertible. Evaluating synthesis at $t = 0$ therefore implies quadrature exactness.

For $P = t^p$, the polynomial $A_p(t-x)$ has leading coefficient $(-1)^p$ as a polynomial in x . Every lower-degree quadrature defect vanishes, so only this coefficient multiplies $E_{p,M}$. The general statement follows by linearity and preservation of leading coefficients. $\square$

9.2 Consequences for phase averages

For any integer $L \geq 1$ and offset α ,

$$\frac{1}{L} \sum_{j=0}^{L-1} \mathcal{Q}_{M,\theta+\alpha+j/L}(P) = \mathcal{Q}_{LM,L(\theta+\alpha)}(P). \quad (9.4)$$

This is simply the bijection $(k, j) \mapsto Lk + j$ between the lattice indices, and holds beyond polynomials whenever the sums exist. The earlier phase-filter manuscript identifies such uniform refinements as the sharp minimum-size filters for uniform translated exactness [3].

Corollary 9.2 (Fixed-phase order of a dyadic refinement). *If $LM = 2^d$, the phase average in (9.4) has exact degree $d + 1$ precisely at the translating phases satisfying*

$$L(\theta + \alpha) \in \mathbb{Z}_{d+1} \pmod{1}.$$

It has exact degree d at every other translating phase. There are exactly $2L$ maximizing phases modulo one, and no translating phase gives degree $d + 2$. If $LM = 2^{r-1}m$ instead, the same conclusion with $d = r - 1$ holds whenever (6.3) or either sufficient cutoff applies.

Proof. Apply the exact refinement identity and the corresponding sharp-order theorem. Multiplication by L on $\mathbb{R}/\mathbb{Z}$ has exactly L preimages of each of the two selected phases. $\square$

This corollary adds a fixed-phase ceiling to an existing uniform-filter classification. It neither changes that classification nor asserts optimality among arbitrary unequal-weight filters at a single phase.

10 Independent exact and numerical verification

The package includes a standalone Python program, `reproducibility/verify.py`. It imports no repository code. Its exact component uses `fractions.Fraction`; its numerical component uses `mpmath` with 60 decimal digits. The two kinds of evidence are recorded separately and are not conflated with the proofs above.

10.1 Exact dyadic evaluator and moment computation

For the bounded Fabius function F on $[0, 1]$, the normalization is $u(x) = F(1 - |x|)$ for $|x| \leq 1$. The classical dyadic recurrences from [2], also used by the earlier repository manuscript, give an exact evaluator. Put

$$d_n = n!2^{n(n-1)/2}F(2^{-n}), \quad d_0 = 1. \quad (10.1)$$

Then

$$(n+1)(2^n-1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k \quad (n \geq 1). \quad (10.2)$$

For a dyadic $x \in (0, 1)$, select n so that $2^{-n} \leq x < 2^{1-n}$ and put $y = x - 2^{-n}$. The identity

$$F(x) = -F(y) + \sum_{k=0}^n \frac{2^{k(k+1)/2}}{k!} F(2^{k-n})y^k \quad (10.3)$$

removes the leading nonzero binary digit. Values at powers of two are computed directly from (10.1), preventing a circular recursive call. The endpoint values are $F(0) = 0$ and $F(1) = 1$.

Moments are computed independently from the distributional identity $Y \stackrel{d}{=} (V + Y')/2$, with V uniform on $[-1, 1]$ and independent of a copy Y' . With $\mu_n = I(x^n)$, odd moments vanish and

$$(2^n - 1)\mu_n = \sum_{\substack{0 \leq k < n \\ k \text{ even}}} \binom{n}{k} \frac{\mu_k}{n - k + 1} \quad (n > 0 \text{ even}), \quad \mu_0 = 1. \quad (10.4)$$

This follows by expanding $(V + Y')^n$ and isolating the last term. The evaluator and moment recurrence therefore approach the quadrature identity from different finite computations.

Table 1: Exact next-degree failures at both maximizing phases. The general theorem, not these finitely many rows, proves all-level maximality.

M	r	Selected phases	First next defect	Second next defect
1	1	0, 1/2	-1/9	5/36
2	2	1/4, 3/4	17/1536	-17/1536
4	3	0, 1/2	97/345600	-799/2764800
8	4	1/4, 3/4	-9919/4529848320	9919/4529848320

The exact run performs 978 assertions. These include 257 reflection checks for F ; lower-degree exactness and first-threshold phase classification on a 16-point phase grid at each mesh 1, 2, 4, 8, 16, 32, 64; next-degree nonvanishing at both selected phases; the four comparison values in Table 1; and 126 exact lattice-refinement identities. Every assertion passed. The complete rational outputs, including the larger fractions through $M = 64$, are in `validation/exact_results.json`.

These exact computations verify finite examples, not infinite-dimensional analytic statements. In particular, testing a finite phase grid does not prove that other phases are absent; Theorem 3.2 supplies that proof.

10.2 Product and derivative diagnostics

The numerical program computes H and H' from 112 product factors and the corresponding logarithmic-derivative sum. For small arguments it uses Taylor expansions to avoid subtracting nearly equal numbers.

It tests both bounds in (3.3) and the derivative bound (5.4) for

$$m \in \{1, 3, 5, 7, 9, 11, 13, 15, 17, 31\}, \quad n \in \{3, 5, \dots, 63\}.$$

These give 930 checks. Two additional checks verify the numerical values of the inequalities for β_1 and D_1 .

For each of the fourteen selected-phase cases through $M = 64$, the program compares the exact rational next defect with (4.3), truncated at odd index 201, and also checks the relative $1/4$ bound. That gives another 28 checks, for 960 numerical assertions in total. All passed. The largest discrepancy in the *normalized* next-defect formula was approximately $2.7651 \cdot 10^{-15}$, and the largest observed relative deviation from its leading term was approximately 0.132868. The latter occurs at the smallest mesh in the test.

These values describe this particular numerical run. They are not rigorous decimal enclosures. The checks compare series discrepancies to an analytic tail envelope, but the arithmetic itself does not use directed rounding. The proofs of the envelope and of the $1/4$ margin are separate. The detailed output is in `validation/numerical_results.json`.

Table 2: Odd-part constants and conservative cutoffs. Decimal constants and the r_* column are numerical diagnostics; the last column follows from the proved elementary formula. For $m = 1$, the sharper dyadic theorem already covers every $r \geq 1$.

m	β_m (approx.)	D_m (approx.)	$r_*(m)$ (approx.)	$2J(m+1) + 4$
1	1.283712	2.264960	5	12
3	5.025370	6.270594	13	28
5	0.338832	14.993681	5	52
7	16.076234	67.714799	35	68
9	-10.887077	148.173749	24	104
13	0.522487	389.713427	8	144
17	-40.763598	7839.404861	84	220
31	121.429034	1302794.266702	245	388

The sufficient inequality itself often succeeds much earlier than either cutoff. In the numerical run, it succeeded at every tested level $r = 1, \dots, 20$ for $m = 1, 3, 5, 7, 9, 11$. For $m = 13, 15, 17$ it succeeded at $r = 3, \dots, 20$, and for $m = 31$ at $r = 7, \dots, 20$. These are observations about a sufficient test, not certified complete classifications of those odd parts. The elementary all-large-level theorem requires no reliance on this table.

10.3 Reproduction and validation boundaries

The exact checks need only Python’s standard library. Run

```
python reproducibility/verify.py -exact-only
```

The numerical checks additionally need `mpmath`:

```
python reproducibility/verify.py -numeric-only
```

Running without either switch performs both sets. The script raises an exception at a failed assertion; it does not silently accept failed tests. The JSON files preserve exact rational strings separately from rounded floating-point displays.

No Lean build was performed. No numerical claim here is described as a formal theorem merely because a Python assertion passed. Conversely, the analytic theorems do not assume a successful test run: their proofs are contained in the preceding sections.

11 Formalization plan and a proof-dependency boundary

11.1 What can be reused

The inspected repository already has the Poisson-defect framework and sufficient parity-phase exactness [5, 4]. Relevant declarations include

```
summable_fourier_monomialRvachevSchwartz
tsum_shifted_monomial_sub_integral
tsum_shifted_monomial_sub_integral_odd
tsum_shifted_polynomial_eq_integral_superconvergent.
```

These names are dependency candidates, not claims that the new results already exist in Lean. The polynomial synthesis layer can reuse the existing deconvolution framework. The classical analytic normalization and zero-order facts should be matched carefully before any new theorem is stated; shifting the Fourier product's starting index changes the support and every mesh factor.

11.2 A staged formal development

A useful first target is Theorem 6.2, without first formalizing the more general odd-part cutoff. The minimal new chain is: obtain the local Taylor coefficient at a zero; split the next defect into odd and even frequencies; prove the derivative envelope with $m = 1$; establish the rational bounds $\beta_1 > 1$ and $D_1 < 5/2$; and combine the summable tails with (6.7).

The proof does not need a full zeta-function library. The bounds $S_3 \leq 7/108$ and $S_4 \leq 1/54$, together with the monotonic comparison used in the proof, suffice for dyadic nonvanishing. Similarly, $\eta_r < 5/8$ suffices for the stationary-point theorem; the decimal constant is not needed.

A second stage can generalize the derivative envelope and prove Theorem 6.1. A third stage can add the integer-valued elementary cutoff. In that final stage it may be cleaner to define J as the least integer at least two satisfying $2^J \geq 2m$, rather than introduce real logarithms and ceilings. This is equivalent to (7.3), but exposes exactly the inequality used by the proof.

The asymptotic expansion should come only after the exact formula and finite tail inequalities. Its useful formal core is not a notation for big- O , but the explicit bound (8.3). Finally, the C^1 perturbation theorem can be proved separately as a general result about a harmonic plus a controlled perturbation.

11.3 Suggested endpoint statements

For dyadic meshes, the natural endpoint is the pair of statements

$$\forall \theta \in Z_{d+1}, \quad E_{d+2,2^d}(\theta) \neq 0,$$

and

$$\forall \theta \in \mathbb{R}/\mathbb{Z}, \quad [Q_{2^d,\theta} \text{ exact on } \mathcal{P}_{d+1}] \iff \theta \in Z_{d+1}.$$

The first is the new maximality content; the second is the already known phase-completeness boundary. Their combination yields (6.5). Working on the circle, or explicitly reducing a real phase modulo one, is essential.

For the general theorem, a formal statement should retain $m > 0$, oddness, $r \geq 1$, and the decomposition $M = 2^{r-1}m$. The sign of A_m and β_m must not be discarded. Using an absolute normalization for the next error would require an extra sign factor and makes the exact identity less transparent; (4.2) deliberately keeps the signed normalization.

12 Further research questions and topics

The results give several concrete directions with explicit success criteria. The following questions are proposed, not solved by the present manuscript.

12.1 Complete the all-integer-mesh maximality classification

Does $E_{r+1,2^{r-1}m}(\theta)$ ever vanish at a phase $\theta \in \mathbb{Z}_r$ for any positive odd m and any $r \geq 1$? The dyadic case is now excluded, and Theorem 7.3 excludes all sufficiently large levels for each fixed odd part. What remains is not a single finite computation, because m is unbounded. A complete theorem would require arithmetic information that is uniform in the odd part, or an exact counterexample.

A practical first milestone is an interval-certified classification for all odd $m \leq B$, checking only the finitely many levels from (7.7). This would turn the new theorem into a rigorous finite-frontier search rather than an arbitrary grid experiment.

12.2 Control near-resonance of the leading coefficient

The obstruction to a one-line extension of the dyadic proof is the possible smallness of $r + \beta_m$. Determine quantitative lower bounds on its distance from zero, or classify its near-integer values. The exact identity suggests that near-resonance may force the $n = 3$ harmonic or the even refined term to become decisive. A proof that these terms cannot simultaneously cancel would go beyond first-harmonic domination.

12.3 Replace the conservative derivative envelope

The term $1/|T(m/2)|$ in D_m can be very large. It results from bounding $|T'(mn/2)|$ by a global support estimate, which ignores the strong arithmetic relation between $mn/2$ and $m/2$. A relative derivative inequality along odd multiples could substantially improve the cutoff. The target is an estimate for $b_{m,n}$ that retains cancellation in the logarithmic derivative rather than separately bounding $P'T$ and PT' .

12.4 Find the optimal dyadic relative margin

The constant $1/4$ in (6.6) is convenient, not sharp. The exact check at $M = 1$, $\theta = 0$ gives a relative deviation around 0.132868 , while the proved upper bound is $47/216 \approx 0.217593$. Is the supremum over all dyadic levels and both selected phases attained at that smallest mesh? Proving such a statement would require a sharper comparison between consecutive levels, not merely separate tail bounds.

12.5 Higher derivatives and the geometry of later defects

The three-factor estimate controls one phase derivative. Additional fixed factors give further powers of odd-multiplier decay, but their residue constants grow. Determine which higher derivative orders admit a uniform harmonic-domination theorem independent of m and r . Separately, classify the zero sets and stationary points of the next and later monomial defects. Their even aliases create a geometry absent from the first defect.

12.6 All-order alias expansions with effective remainders

Retaining any finite set of odd frequencies in (4.3) gives additional exponential scales. For higher monomial degrees, several valuation layers survive. Derive a systematic expansion indexed jointly by frequency and its two-adic valuation, with explicit coefficients from local Taylor jets of H and uniform truncation bounds. The present $2^{-r}/3^{-r}$ split is the first nontrivial case of that program.

12.7 Generalized geometric convolution kernels

Replace the dyadic uniform series by other geometrically weighted convolutions. Which arithmetic properties of their Fourier zeros permit complete phase classification and fixed-phase ceilings? Integer bases larger than two have more residue classes and more candidate phases. One should distinguish what follows from zero multiplicities alone from what needs special sine-ratio estimates like (3.3).

12.8 Certified filters robust to phase and weight errors

The perturbation theorem controls the normalized first defect when a concrete implementation can bound its C^1 perturbation. Develop explicit bounds translating errors in sampled values, phase locations, and filter weights into those norms. Combining such bounds with the next-degree nonvanishing margin would provide a certified region in which both the number of superconvergent phases and the fixed-phase ceiling survive. This is a stronger numerical-design question than merely observing stable plots.

13 Conclusion

The repository’s first-superconvergence phase classification leaves a logically separate question: whether a special phase might also cancel the next monomial error. The exact odd/even decomposition answers that question at every dyadic mesh, with a uniform relative nonvanishing margin. The same mechanism produces effective eventual maximality for every fixed odd part, including a completely elementary finite-level reduction.

The proof also explains the geometry and scale separation of the errors. A third sinc factor controls the derivative of the normalized first defect, while the next defect divides into a same-mesh odd series and a refined first defect. These two observations supply a foundation for further exact classification, certified computation, and formalization, without treating numerical checks as proofs or silently extending a dyadic statement to all integer meshes.

A A compact dependency and novelty ledger

Result or ingredient	Status and role
Fourier product and zero multiplicities	Classical kernel structure; Section 2 fixes normalization and supplies a direct proof of the part used here.
Two-factor multiplier bound and first-phase completeness	Present in [3]; re-established as the baseline, not claimed new.
Three-factor multiplier bound and complete stationary-point geometry	Developed here in Lemma 3.1 and Theorem 3.2; gives a relative C^1 control, rather than only the first zero set.
Exact next-defect decomposition	Developed in Theorem 4.2; keeps the derivative term and the entire refined even-frequency component.
Sharp dyadic fixed-phase degree	Main completion in Theorem 6.2; covers every level, both selected phases, and gives a uniform signed relative margin.
General odd-part maximality	Sufficient criterion and eventual theorem in Theorems 6.1 and 7.1; not an unconditional all-level theorem for every odd m .
Elementary finite-level reduction	Theorem 7.3 gives the explicit $O(m \log(m + 1))$ cutoff without evaluating a special-function constant.
Parity-dependent exponential expansion	Theorem 8.1 gives exact retained terms and a finite-error inequality.

Result or ingredient	Status and role
Synthesis and uniform-grid refinement	Existing frameworks; Theorem 9.1 and Corollary 9.2 transfer the new sharp ceilings.
Formal verification	No new Lean theorem or Lean build is claimed. The proof boundary and staged plan are given in Section 11.

B Normalization and edge-case checklist

The first Fourier factor is $\text{sinc}(\pi t)$, not $\text{sinc}(\pi t/2)$. The support is $[-1, 1]$, the integral is one, and $u(0) = 1$. The latter follows by convolving the first uniform summand, whose density equals one on $[-1/2, 1/2]$, with the remaining law supported on that interval. It is also part of the classical normalization.

The case $M = 1$ is included: $r = 1$ and the base exact degree is zero. The phase circle identifies 0 with 1; no theorem excludes integer translates of a selected representative. The odd part is positive; negative frequencies are handled by evenness of H , not by treating m as signed. The factor A_m is nonzero but may be negative. Every next-error formula retains its sign. The integer cutoff uses $r = d + 1$, so a cutoff on r becomes a cutoff one smaller on d .

Finally, exactness through degree s is a simultaneous statement about all polynomials of degree at most s . A single higher monomial can sometimes be integrated accidentally even when lower ones fail. Such an isolated equality does not increase the exact degree used in this article.

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